Containerization-and-DevOps

🐳 Class Task

🛠️ Tools Used

- Docker
- WSL
- Ubuntu

⚙️ Steps

Create a Docker Volume

```
docker volume create myvolume
```

List Volumes

```
docker volume ls
```

Run Container with Volume

```
docker run -it --name test -v test:/home/app ubuntu /bin/bash
```

Create Directory and File inside Container

```
mkdir -p /home/app
cd /home/app
echo "sapid123" > sapid.txt
```

Exit and Check Containers

```
exit
docker ps -a
```

Remove Container

```
docker rm <container_id>
```

Run Container Again with Volume

```
docker run -it --name test2 -v test:/home/app ubuntu /bin/bash
```

Verify File

```
cat /home/app/sapid.txt
```

(File not found due to incorrect volume mapping)

Run Container with Correct Bind Mount

```
docker run -it --rm -v ./test:/home/app ubuntu /bin/bash
```

Create File Again

```
cd /home/app
echo test1234321 > sapid.txt
exit
```

Verify from Host System

```
cd test
cat sapid.txt
```

⚠️ Notes

- Named volume (`test`) did not map as expected
- Bind mount (`./test:/home/app`) worked correctly
- Data persisted between container and host

✅ Outcome

- Created and used Docker volumes
- Understood difference between named volume and bind mount
- Verified data persistence

📸 Screenshots

```
PS C:\Users\srist> wsl
wsl: Unknown key 'wsl2.systemd' in C:\Users\srist\.wslconfig:2
sristee@Sristee:/mnt/c/Users/srist$ docker volume create myvolume
myvolume
sristee@Sristee:/mnt/c/Users/srist$ docker volume ls
DRIVER    VOLUME NAME
local     myvolume
sristee@Sristee:/mnt/c/Users/srist$ docker run -it --name test -v test/home/app ubuntu /bin/bash
root@918b15bd87a9:/# cd /home/app
bash: cd: /home/app: No such file or directory
root@918b15bd87a9:/# cd/home/app
bash: cd/home/app: No such file or directory
root@918b15bd87a9:/# mkdir -p /home/app
cd /home/app
root@918b15bd87a9:/home/app# echo "sapid123' > sapid.txt
> sapid.txt
>
>
> ^C
root@918b15bd87a9:/home/app# exit
exit
sristee@Sristee:/mnt/c/Users/srist$ docker ps -a
CONTAINER ID   IMAGE                 COMMAND                  CREATED          STATUS                        PORTS
                                NAMES
918b15bd87a9   ubuntu                "/bin/bash"              15 minutes ago   Exited (130) 10 seconds ago
                                test
5b1a1944aa35   nginx                 "/docker-entrypoint.…"   4 days ago       Exited (255) 4 days ago       0.0.0.0:808
0->80/tcp, [::]:8080->80/tcp    nginx-container
7d9d844949a8   8971fdfc5261          "java Hello"             11 days ago      Exited (0) 11 days ago
                                sweet_franklin
```

```
5558972a33c1   8971fdfc5261          "java Hello"             12 days ago      Exited (0) 12 days ago
                                gallant_lederberg
d7de2b318b20   myrepo/java-app:1.0   "bash"                   2 weeks ago      Exited (0) 2 weeks ago
                                charming_cartwright
d021472c9ac2   myrepo/java-app:1.0   "bash"                   2 weeks ago      Exited (127) 2 weeks ago
                                blissful_dewdney
b3061d2bb92f   ubuntu                "bash"                   2 weeks ago      Exited (127) 2 weeks ago
                                javac
b92d4acfaec0   hello-world           "/hello"                 2 weeks ago      Exited (0) 2 weeks ago
                                goofy_aryabhata
715286b13da5   ubuntu                "/bin/bash"              2 weeks ago      Exited (0) 2 weeks ago
                                quirky_saha
378280b9d749   ubuntu                "/bin/bash"              2 weeks ago      Exited (0) 2 weeks ago
                                upbeat_stonebraker
c7cd6ccf7635   ubuntu                "/bin/bash"              2 weeks ago      Exited (0) 2 weeks ago
                                frosty_rubin
bd846f5d38ab   ubuntu                "/bin/bash"              2 weeks ago      Exited (0) 2 weeks ago
                                distracted_johnson
46e5a7258c51   ubuntu                "/bin/bash"              2 weeks ago      Exited (0) 2 weeks ago
                                upbeat_shockley
sristee@Sristee:/mnt/c/Users/srist$ docker rm 9bf
Error response from daemon: No such container: 9bf
sristee@Sristee:/mnt/c/Users/srist$ docker rm 46e
46e
sristee@Sristee:/mnt/c/Users/srist$ docker run -it --name test2 -v test/home/app ubuntu /bin/bash
root@12b831035f59:/# ls
bin  boot  dev  etc  home  lib  lib64  media  mnt  opt  proc  root  run  sbin  srv  sys  test  tmp  usr  var
root@12b831035f59:/# cat /home/app/sapid.txt
cat: /home/app/sapid.txt: No such file or directory
```

```
root@12b831035f59:/# cd /home/app/sapid.txt
bash: cd: /home/app/sapid.txt: No such file or directory
root@12b831035f59:/# exit
exit
sristee@Sristee:/mnt/c/Users/srist$ docker run -it --name test3 -v test/home/app ubuntu /bin/bash
root@f83fe34b7d69:/# ls /home
ubuntu
root@f83fe34b7d69:/# cd home
root@f83fe34b7d69:/home# cd app
bash: cd: app: No such file or directory
root@f83fe34b7d69:/home# ^C
root@f83fe34b7d69:/home# exit
exit
sristee@Sristee:/mnt/c/Users/srist$ docker run -it --rm -v ./test:/home/app ubuntu /bin/bash
root@7c5b409c223a:/# cd /home/app/
root@7c5b409c223a:/home/app# echo test1234321 > sapid.txt
root@7c5b409c223a:/home/app# exit
exit
sristee@Sristee:/mnt/c/Users/srist$ ls
 AppData                                                                    Recent
'Application Data'                                                         'Saved Games'
 Contacts                                                                   Searches
 Cookies                                                                    SendTo
 CrossDevice                                                               'Start Menu'
 Dockerfile                                                                 Templates
 Documents                                                                  Vagrantfile
 Downloads                                                                  Videos
 Favorites                                                                 'VirtualBox VMs'
 Links                                                                      container.tar
'Local Settings'                                                            image-processing-app
```

```
 Music                                                                      image.tar
'My Documents'                                                              java-app.tar
 NTUSER.DAT                                                                 java-docker
 NTUSER.DAT{b637632a-22a0-11f0-8b4a-5802054d40cd}.TM.blf                    lambda-layer
 NTUSER.DAT{b637632a-22a0-11f0-8b4a-5802054d40cd}.TMContainer00000000000000000001.regtrans-ms   lambda_function.py
 NTUSER.DAT{b637632a-22a0-11f0-8b4a-5802054d40cd}.TMContainer00000000000000000002.regtrans-ms   ntuser.dat.LOG1
 NetHood                                                                    ntuser.dat.LOG2
 OneDrive                                                                   ntuser.ini
 PrintHood                                                                  test
sristee@Sristee:/mnt/c/Users/srist$ cd test
sristee@Sristee:/mnt/c/Users/srist/test$ cat sapid.txt
test1234321
sristee@Sristee:/mnt/c/Users/srist/test$
```